Chapter 16

Integral Calculus

16

SYLLABUS 5.5 5.9 5.10 5.11 5.15 5.16 5.17

Your car's speedometer tells you how fast you are going at every instant. Suppose you record it for an entire journey. Can you work out how far you travelled, using only the speed?

Over a tiny slice of time, your speed barely changes, so the distance covered in that slice is roughly speed times the slice of time. Add up all the slices and you have the total distance. As the slices shrink toward zero width, the rough sum becomes exact. What you are computing is the area under the speed-time graph.

This is the second great idea of calculus, and it arrives as two ideas rather than one. The first is **reversing a derivative**: given a rate of change, recover the quantity it came from. The second is **accumulating**: adding up a great many thin slices to get a total. The area under a graph is exactly this kind of total.

They have no obvious connection. Reversing a derivative is a problem in algebra. Adding slices is a problem in geometry. Yet the car journey links them, because the distance travelled is both the reverse of the velocity and the area under the velocity graph.

The chapter is built in that order. Section 16.1 does the reversal on its own, with no mention of area. Section 16.2 defines the area on its own, with no mention of derivatives, and then proves that the two are the same calculation. That result is the Fundamental Theorem of Calculus, and once it is in place the accumulated totals, areas between curves, volumes of revolution and distances travelled all become routine.

16.1 Antiderivatives

This chapter runs differentiation backwards. Given a function f, we look for a function whose derivative is f. Such a function is called an **antiderivative** of f.

There is never just one. The derivative of a constant is zero, so a constant can be added to any antiderivative without changing its derivative. Antiderivatives therefore come in families, all differing by a constant.

DEF F is an **antiderivative** of f when $F'(x) = f(x)$. If F is one antiderivative, then $F(x) + C$ is an antiderivative for every constant C, and these are all of them.

Reversing the power rule means raising the power by one and dividing by the new power.

KEY · IB An antiderivative of x^n is $\dfrac{x^{n+1}}{n+1} + C$, for $n \neq -1$.

The exception $n = -1$ is handled separately in the next section. Constant multiples and sums are reversed term by term, just as they are differentiated term by term.

EXAMPLE 16.1

Find the antiderivatives of $3x^2 - 4x + 1$.
Reverse each term, raising each power by one:

$$F(x) = x^3 - 2x^2 + x + C.$$

Check by differentiating: $\frac{d}{dx}(x^3 - 2x^2 + x + C) = 3x^2 - 4x + 1$. Differentiating your answer costs nothing and checks it completely.

Boundary conditions

The $+C$ means an antiderivative is not unique until you pin it down. One extra piece of information, a point the curve passes through, fixes C and selects a single function.

EXAMPLE 16.2

A curve has gradient $\dfrac{dy}{dx} = 6x^2 - 2$ and passes through $(1, 4)$. Find its equation.
Antidifferentiate to recover y:

$$y = 2x^3 - 2x + C.$$

Use the point $(1, 4)$: $4 = 2 - 2 + C \implies C = 4$. So $y = 2x^3 - 2x + 4$.

A symbol for the family

Writing "the antiderivatives of f" every time is slow, so there is a symbol for it.

KEY

$$\int f(x)\,dx = F(x) + C \qquad \text{means} \qquad F'(x) = f(x).$$

This is called the **indefinite integral** of f, and the C is the **constant of integration**. It is easy to leave out, and an answer written without it names one antiderivative instead of all of them.

For now the symbol is nothing more than shorthand: $\int f(x)\,dx$ means the family of antiderivatives, and no other meaning has been given to it. Its shape is worth noticing, though. The sign is an elongated letter S, and the S stands for *sum*. Nothing in this section has involved a sum of any kind, and that is deliberate. In §16.2 we shall define a completely separate quantity, the area under a curve, as a limit of sums, and only then prove that the two coincide. Until that proof, read the symbol as shorthand and nothing more.

YOUR TURN 16.1 Antiderivatives

1. Find the antiderivatives of each function.

(a) $4x^3 - 6x$ [1] (b) $x^2 + 3x - 5$ [2]

(c) $\dfrac{6}{x^3}$ [2] (d) $\sqrt{x}$ [2]

(e) $x^2 - \dfrac{1}{x^2}$ [2] (f) $\dfrac{x^3 - 2x}{x}$ [3]

2.

(a) A curve has $\dfrac{dy}{dx} = 4x - 3$ and passes through $(2, 5)$. Find y. [3]

(b) A curve has $\dfrac{dy}{dx} = 3\sqrt{x}$ and passes through $(4, 20)$. Find y. [4]

(c) Explain why a gradient function alone does not determine a curve, and what extra information does. [2]

16.2 Definite Integrals and Area

Set antiderivatives aside for the moment. This section starts a second, unrelated problem, and only later do the two meet.

The problem is area. For a region with straight edges there are formulas, but under a curve there are none, because the boundary is different at every point. What can be done is to approximate.

Cut the interval from a to b into n strips, each of width Δx. Replace each strip by a rectangle whose height is the value of the function somewhere in that strip. Every rectangle has area height $\times$ width, so the total is

$$\sum_{i=1}^{n} f(x_i)\,\Delta x.$$

This is not the area. It is a sum of rectangles, and it is wrong by however much the rectangles miss or overshoot. But narrower strips leave less room for error, and as n grows the sum settles on a single number.

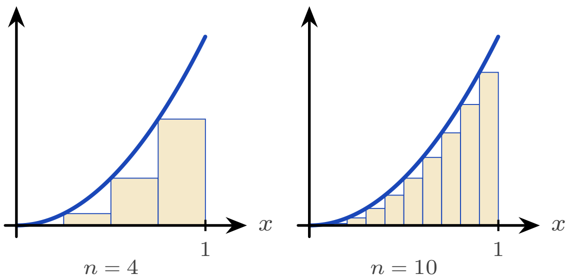

Figure 16.1 Rectangles under $y = x^2$ on $[0, 1]$, taking the height at the left of each strip. Four strips give 0.219 and ten give 0.285; the true area is $\frac{1}{3}$.

The numbers show it happening. Taking the height at the left of each strip underestimates the area, and taking it at the right overestimates it, so the true value is caught between them.

n	4	10	100	1000
low estimate	0.219	0.285	0.328	0.3328
high estimate	0.469	0.385	0.338	0.3338

The two columns close on each other, and the number they trap is $\frac{1}{3}$. That limit is what we mean by the area, and it is called the **definite integral**.

DEF The **definite integral** of f from a to b is the limit of the rectangle sums as the strips become arbitrarily narrow:

$$\int_a^b f(x)\,dx = \lim_{\Delta x \to 0} \sum_i f(x_i)\,\Delta x.$$

For $f(x) \geq 0$ it is the area between the curve and the x-axis from a to b.

Now the symbol makes sense. The elongated S is a sum, $f(x)\,dx$ is one rectangle of height $f(x)$ and vanishing width dx, and the integral adds them. That is what the notation was built to say.

It is also, so far, useless for calculation. Nothing here tells you how to find such a limit, and nobody wants to add a thousand rectangles by hand.

Why the two ideas agree

The way out is surprising. This limit of sums was defined without mentioning derivatives at all, yet it can be found by antidifferentiation. Here is why.

Fix the left end a and let the right end move. Write $A(x)$ for the area under the curve from a to x, so that A is a function of where you stop. Now ask how A changes when x increases by a small amount h. The extra area, $A(x+h) - A(x)$, is a single thin strip of width h.

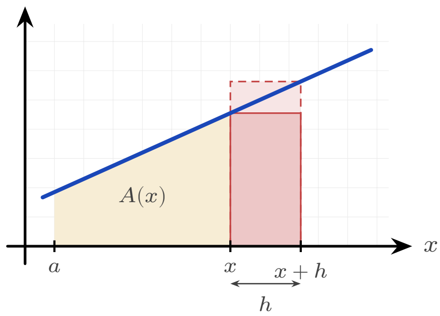

Figure 16.2 The extra area $A(x+h) - A(x)$ is one thin strip. It contains the short rectangle and is contained in the tall one, so its area lies between h times the smallest height of f on the strip and h times the largest.

That strip is trapped. It contains a rectangle of width h whose height is the *smallest* value of f on the strip, and it fits inside a rectangle of width h whose height is the *largest* value. Writing m and M for those two heights,

$$m\,h \;\leq\; A(x+h) - A(x) \;\leq\; M\,h.$$

Divide through by h:

$$m \leq \frac{A(x+h) - A(x)}{h} \leq M.$$

Now let $h \to 0$. The strip shrinks to the single point x, so its smallest and largest heights both close on $f(x)$. The quantity in the middle is squeezed between them, and it is exactly the definition of $A'(x)$. Therefore

$$A'(x) = f(x).$$

That is the whole argument, and it gives the result the chapter was built to reach: **the area function is an antiderivative of f.**

The computational rule follows in two lines. If F is any antiderivative of f, then A and F differ by a constant, so $A(x) = F(x) + C$. Putting $x = a$ gives $0 = F(a) + C$, since the area from a to a is nothing, so $C = -F(a)$. Putting $x = b$ then gives the area from a to b.

KEY **The Fundamental Theorem of Calculus.**

$$\int_a^b f(x)\,dx = \Big[F(x)\Big]_a^b = F(b) - F(a), \qquad \text{where } F' = f.$$

So the two ideas are one calculation. Reversing a derivative and adding up rectangles give the same answer. The constant C cancels in the subtraction, which is why a definite integral is a plain number while an indefinite one carries a $+C$.

KEY · IB

$$A = \int_a^b y\,dx \qquad (y \geq 0)$$

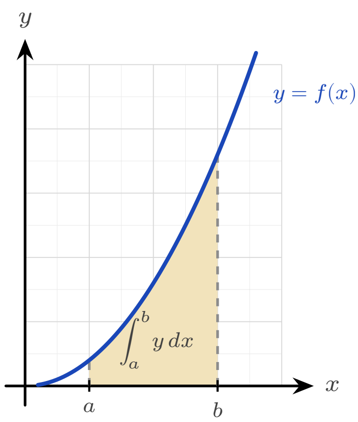

Figure 16.3 For $y \geq 0$ the definite integral measures the shaded area between the curve and the x-axis, from $x = a$ to $x = b$.

EXAMPLE 16.3

Evaluate $\int_0^2 3x^2\,dx$.

$$\int_0^2 3x^2\,dx = \left[x^3\right]_0^2 = 8 - 0 = 8.$$

TIP *Your GDC evaluates definite integrals directly, and on a calculator paper that is often all a question requires. Even when analytic working is demanded, the GDC value is a free check of your antiderivative.*

Areas below the axis

Where the curve dips below the x-axis, the integral is negative: it is a signed area. For a true geometric area you integrate the parts separately and take the absolute value of any negative piece.

EXAMPLE 16.4

Find the area enclosed between $y = x^2 - 4$ and the x-axis from $x = 0$ to $x = 2$.

On $[0, 2]$ the curve lies below the axis, so the integral is negative:

$$\int_0^2 (x^2 - 4)\,dx = \left[\tfrac{x^3}{3} - 4x\right]_0^2 = \tfrac{8}{3} - 8 = -\tfrac{16}{3}.$$

The area is the magnitude, $\dfrac{16}{3}$. An area is never negative, so $-\frac{16}{3}$ answers a different question from the one asked.

CAUTION *Before finding an area, check where the curve crosses the x-axis. A curve that goes above and below will have positive and negative parts that partly cancel in a single integral, giving the wrong area. Split at the roots.*

YOUR TURN 16.2 Definite Integrals and Area

3. Evaluate each definite integral.

(a) $\displaystyle\int_1^2 4x\,dx$ [2]

(b) $\displaystyle\int_0^3 (x^2 + 1)\,dx$ [2]

(c) $\displaystyle\int_1^e \frac{1}{x}\,dx$ [2]

(d) $\displaystyle\int_1^4 \frac{1}{\sqrt{x}}\,dx$ [2]

4.

(a) Find the area under $y = x^2$ between $x = 1$ and $x = 3$. [2]

(b) Find the area enclosed between $y = x^2 - 1$ and the x-axis. [3]

(c) Find the area under $y = \sin x$ from $x = 0$ to $x = \pi$. [2]

(d) Evaluate $\int_0^{2\pi} \sin x \, dx$, and explain why the answer is not the area between the curve and the axis over that interval. [3]

5. The area under $y = x^2$ from $x = 0$ to $x = 2$ is to be estimated by four rectangles of equal width.

(a) Taking the height of each rectangle at the left of its strip, show that the estimate is 1.75. [3]

(b) Taking the height at the right instead, show that the estimate is 3.75. [2]

(c) Find the exact area, and confirm that it lies between the two estimates. [3]

(d) State what happens to the two estimates as the number of rectangles increases. [2]

16.3 Standard Integrals

Every derivative you learned has a reverse. Reading the derivative table backwards gives the standard integrals.

KEY · IB

$$\int \frac{1}{x}\, dx = \ln|x| + C, \qquad \int \sin x \, dx = -\cos x + C, \qquad \int \cos x \, dx = \sin x + C, \qquad \int e^x \, dx = e^x + C$$

The first fills the gap left by the power rule at $n = -1$: the antiderivative of x^{-1} is not a power but a logarithm. The absolute value lets it work for negative x too.

At HL, three more standard integrals are provided, the reverses of the exponential and inverse-trigonometric derivatives.

KEY · IB HL

$$\int a^x \, dx = \frac{a^x}{\ln a} + C, \qquad \int \frac{1}{a^2 + x^2}\, dx = \frac{1}{a}\arctan\frac{x}{a} + C, \qquad \int \frac{1}{\sqrt{a^2 - x^2}}\, dx = \arcsin\frac{x}{a} + C$$

Reading the rest of the HL derivative table backwards gives four more. They are worth learning, because the formula booklet gives only the derivatives:

KEY

$$\int \sec^2 x\,dx = \tan x + C, \qquad \int \sec x \tan x\,dx = \sec x + C,$$
$$\int \operatorname{cosec}^2 x\,dx = -\cot x + C, \qquad \int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x + C$$

Linear composites

Sometimes the function inside is linear, as in e^{2x} or $\cos 3x$. Integrate as usual, then divide by the coefficient of x inside. This reverses the chain rule, whose extra factor was that same coefficient.

KEY

$$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C, \qquad \text{where } F' = f$$

EXAMPLE 16.5

Find (a) $\int e^{2x}\,dx$, (b) $\int (2x+1)^5\,dx$, (c) $\int \cos 3x\,dx$.
Each has a linear function inside, so integrate and then divide by the coefficient of x:

$$\text{(a) } \tfrac{1}{2}e^{2x} + C, \qquad \text{(b) } \frac{(2x+1)^6}{12} + C, \qquad \text{(c) } \tfrac{1}{3}\sin 3x + C.$$

In (b), reversing the power rule gives $\frac{(2x+1)^6}{6}$, then dividing by the coefficient 2 gives the 12.

EXAMPLE 16.6

Find (a) $\int \frac{1}{x^2+4}\,dx$, (b) $\int \frac{1}{\sqrt{9-x^2}}\,dx$.
Match the standard forms with $a = 2$ and $a = 3$:

$$\text{(a) } \tfrac{1}{2}\arctan\frac{x}{2} + C, \qquad \text{(b) } \arcsin\frac{x}{3} + C.$$

The only work is recognising a^2 as 4 and as 9. After that the standard form gives the answer.

The linear-composite rule extends to the whole table. For instance, $\int \sec^2(2x+5)\,dx = \frac{1}{2}\tan(2x+5) + C$, the guide's own example: integrate as usual, divide by the inner coefficient.

YOUR TURN 16.3 Standard Integrals

6. Find each indefinite integral.

(a) $\displaystyle\int (2\cos x - \sin x)\, dx$ [2]

(b) $\displaystyle\int \left(3e^x + \frac{1}{x}\right) dx$ [2]

(c) $\displaystyle\int e^{3x}\, dx$ [2]

(d) $\displaystyle\int (3x-2)^4\, dx$ [2]

(e) $\displaystyle\int \sin 2x\, dx$ [2]

(f) $\displaystyle\int \sec^2(4x)\, dx$ [2]

7. GDC Find each indefinite integral, matching it to a standard form.

(a) $\displaystyle\int \frac{1}{x^2+9}\, dx$ [2]

(b) $\displaystyle\int \frac{1}{\sqrt{16-x^2}}\, dx$ [2]

(c) $\displaystyle\int 5^x\, dx$ [2]

(d) $\displaystyle\int \frac{1}{x^2+4x+13}\, dx$ [4]

16.4 Rewriting Before Integrating HL

Many integrands do not match a standard form as they stand, but can be rewritten until they do. Three rewritings cover almost every case: divide, complete the square, or split into partial fractions. Deciding between them takes one look at the denominator.

Divide first

When the numerator has degree at least that of the denominator, the fraction is **improper** and no standard form applies. Divide before anything else. For instance $\dfrac{x^2}{x+1} = x - 1 + \dfrac{1}{x+1}$, so the integral is $\frac{x^2}{2} - x + \ln|x+1| + C$. Only when the numerator has the lower degree are the next two methods available.

Completing the square HL

An examination question rarely presents a denominator already in the form a^2+x^2. A quadratic such as x^2+2x+5 has that shape, but only after you complete the square.

EXAMPLE 16.7

Find $\displaystyle\int \frac{1}{x^2+2x+5}\, dx$.

Complete the square: $x^2+2x+5 = (x+1)^2+4$. This is the arctangent shape with $a=2$ and the shifted variable $x+1$:

$$\int \frac{dx}{(x+1)^2+4} = \frac{1}{2}\arctan\left(\frac{x+1}{2}\right) + C.$$

Any quadratic with no real roots can be written in the form a^2+u^2 this way. The shift from x to $x+1$ does not change the integral, because the derivative of $x+1$ is 1.

Partial fractions HL

When the quadratic *does* factorise, use a different method. Split the fraction into partial fractions, as in Ch. 6. Each piece then has the form $\frac{1}{x+k}$, and integrates to a logarithm.

EXAMPLE 16.8

Find $\displaystyle\int \frac{1}{x^2+3x+2}\,dx$.

Factor and decompose: $\dfrac{1}{(x+1)(x+2)} = \dfrac{1}{x+1} - \dfrac{1}{x+2}$. Then

$$\int \left(\frac{1}{x+1} - \frac{1}{x+2}\right) dx = \ln|x+1| - \ln|x+2| + C = \ln\left|\frac{x+1}{x+2}\right| + C.$$

Factorisable denominator: partial fractions and logarithms. Irreducible denominator: complete the square and arctangent. Choosing between the two is the first decision, and factorising the denominator is what settles it.

YOUR TURN 16.4 Rewriting Before Integrating HL

8.

(a) Find $\displaystyle\int \frac{1}{x^2-9}\,dx$ using partial fractions. [4]

(b) Explain how you decide, from the denominator alone, whether to use partial fractions or to complete the square. [2]

16.5 Integration by Substitution

Substitution reverses the chain rule. Look for a composite function whose inner function also appears, up to a constant multiple, as a factor outside. Put u equal to that inner function, and a hard integral in x becomes an easy one in u.

The method has four steps. Let u be the inner function. Replace dx using $du = \dfrac{du}{dx}\,dx$. Integrate in u. Then put x back.

EXAMPLE 16.9

Find $\displaystyle\int 2x(x^2+1)^3\,dx$.

Let $u = x^2+1$, so $\dfrac{du}{dx} = 2x$, giving $du = 2x\,dx$. The integral becomes

$$\int u^3\,du = \frac{u^4}{4} + C = \frac{(x^2+1)^4}{4} + C.$$

The factor $2x$ outside is exactly the $\frac{du}{dx}$ required. Without it the substitution would leave an x

behind and fail.

EXAMPLE 16.10

Find $\displaystyle\int \frac{2x}{x^2+1}\,dx$.

With $u = x^2+1$, $du = 2x\,dx$:

$$\int \frac{1}{u}\,du = \ln|u| + C = \ln(x^2+1) + C.$$

An integral of the form $\dfrac{f'(x)}{f(x)}$ always integrates to $\ln|f(x)|$.

CAUTION *For a definite integral by substitution, either convert the limits to u-values, or substitute back to x before applying the original limits. Applying the x-limits to a u-expression gives a wrong answer.*

EXAMPLE 16.11

Evaluate $\displaystyle\int_0^{\sqrt{3}} x\sqrt{x^2+1}\,dx$.

Let $u = x^2+1$, so $du = 2x\,dx$. Convert the limits too: $x = 0 \Rightarrow u = 1$ and $x = \sqrt{3} \Rightarrow u = 4$. Then

$$\int_0^{\sqrt{3}} x\sqrt{x^2+1}\,dx = \frac{1}{2}\int_1^4 \sqrt{u}\,du = \frac{1}{2}\cdot\frac{2}{3}\Big[u^{3/2}\Big]_1^4 = \frac{1}{3}(8-1) = \frac{7}{3}.$$

Once the limits are converted, you never return to x at all; the whole calculation finishes in u.

YOUR TURN 16.5 Integration by Substitution

9. Find each integral by substitution.

(a) $\displaystyle\int 3x^2(x^3+1)^4\,dx$ [2]

(b) $\displaystyle\int xe^{x^2}\,dx$ [2]

(c) $\displaystyle\int 2x\sqrt{x^2+1}\,dx$ [3]

(d) $\displaystyle\int \frac{x}{(x^2+4)^2}\,dx$ [3]

10.

(a) Evaluate $\displaystyle\int_0^1 3x^2(x^3+1)^4\,dx$ by converting the limits. [3]

(b) Evaluate the same integral by substituting back to x first, and confirm the answers agree. [3]

16.6 Integration by Parts HL

Substitution reverses the chain rule; **integration by parts** reverses the product rule. It handles products that substitution cannot, such as xe^x or $\ln x$.

KEY · IB

$$\int u\,\frac{dv}{dx}\,dx = uv - \int v\,\frac{du}{dx}\,dx$$

You split the integrand into a part u to differentiate and a part $\frac{dv}{dx}$ to integrate. Choose u to be the part that gets simpler when differentiated, so the new integral is easier than the original.

EXAMPLE 16.12

Find $\displaystyle\int xe^x\,dx$.

Let $u = x$ (simplifies to 1) and $\frac{dv}{dx} = e^x$ (so $v = e^x$):

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C = e^x(x-1) + C.$$

Choosing $u = x$ is what makes the new integral, $\int e^x\,dx$, trivial. The other choice would make it worse.

EXAMPLE 16.13

Find $\displaystyle\int \ln x\,dx$.

There is only one function, but write $u = \ln x$ and $\frac{dv}{dx} = 1$ (so $v = x$):

$$\int \ln x\,dx = x\ln x - \int x\cdot\frac{1}{x}\,dx = x\ln x - x + C.$$

Taking $\frac{dv}{dx} = 1$ is worth remembering. It is how both $\ln x$ and $\arctan x$ are integrated.

Repeated integration by parts

If one application of parts leaves an integral that still needs parts, apply it again. Each round of differentiation reduces the power of x by one.

EXAMPLE 16.14

Find $\displaystyle\int x^2e^x\,dx$.

First round, $u = x^2$:

$$\int x^2e^x\,dx = x^2e^x - 2\int xe^x\,dx.$$

The remaining integral is the earlier example, $\int xe^x\,dx = e^x(x-1)$:

$$\int x^2e^x\,dx = x^2e^x - 2e^x(x-1) + C = e^x(x^2-2x+2) + C.$$

A power x^n takes n rounds of parts to clear, since only the power changes from round to round; the exponential returns unchanged each time.

Something cleverer happens when *neither* factor ever simplifies. For $e^x \sin x$, two rounds of parts bring the original integral back, and you solve for it like an equation.

EXAMPLE 16.15

Find $I = \displaystyle\int e^x \sin x\,dx$.

Parts with $u = \sin x$:

$$I = e^x \sin x - \int e^x \cos x\,dx.$$

Parts again on the new integral, $u = \cos x$:

$$\int e^x \cos x\,dx = e^x \cos x + \int e^x \sin x\,dx = e^x \cos x + I.$$

Substitute back:

$$I = e^x \sin x - e^x \cos x - I \implies 2I = e^x(\sin x - \cos x) \implies I = \tfrac{1}{2}e^x(\sin x - \cos x) + C.$$

The original integral reappeared on the right-hand side, which turned the problem into an equation for I. Watch for this whenever neither factor gets simpler: two rounds of parts bring I back, and algebra finishes the job.

Choosing a method

Five methods are now available, and most marks lost in this topic are lost by starting the wrong one. The choice is settled by looking at the integrand before writing anything.

What you see	Method	Example
A standard form exactly, or with a linear inside	read it off, dividing by the inner coefficient	$\cos 3x$, e^{2x}, $(2x+1)^5$
A function together with a multiple of its own derivative	substitution	$2x(x^2+1)^3$, $\dfrac{2x}{x^2+1}$
A product of two unlike functions, one of which simplifies when differentiated	parts	xe^x, $x\sin x$, $\ln x$
A fraction whose denominator factorises	partial fractions, then logarithms	$\dfrac{1}{x^2+3x+2}$
A fraction whose denominator is a quadratic with no real roots	complete the square, then arctangent	$\dfrac{1}{x^2+2x+5}$

Work down the list in that order, because the earlier tests are quicker to apply and the earlier methods are shorter. Two habits help further. Check first whether the fraction is improper, since dividing may remove the problem entirely. And if a substitution leaves an integral no easier than the one you started with, the substitution was the wrong one; go back rather than pressing on.

YOUR TURN 16.6 Integration by Parts HL

11. Find each integral by parts.

(a) $\displaystyle\int x\cos x\,dx$ [3]

(b) $\displaystyle\int xe^{2x}\,dx$ [3]

(c) $\displaystyle\int x\ln x\,dx$ [3]

(d) $\displaystyle\int \arctan x\,dx$ [4]

12.

(a) Evaluate $\displaystyle\int_0^{\pi} x\sin x\,dx$. [3]

(b) Evaluate $\displaystyle\int_0^{1} xe^x\,dx$. [3]

(c) State which factor you chose as u in part (b), and why the other choice would not help. [2]

16.7 Areas Between Curves

To find the area between two curves, integrate the difference: the upper curve minus the lower. The signed-area problem disappears, because the difference is positive wherever the first curve is on top, regardless of the axis.

KEY

$$A = \int_a^b (y_{\text{top}} - y_{\text{bottom}})\, dx$$

The limits a and b are usually the x-coordinates where the curves intersect, found by setting them equal.

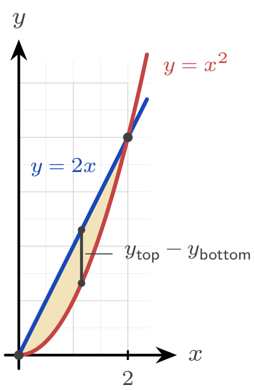

Figure 16.4 The region enclosed between $y = 2x$ and $y = x^2$. The curves meet where $2x = x^2$, at $x = 0$ and $x = 2$, and the line is the upper boundary in between, so the area is $\int_0^2 (2x - x^2)\, dx$.

EXAMPLE 16.16

Find the area enclosed between $y = 2x$ and $y = x^2$.

Intersections: $2x = x^2 \Rightarrow x = 0$ or $x = 2$. On $(0, 2)$ the line $y = 2x$ is above, so

$$A = \int_0^2 (2x - x^2)\, dx = \left[x^2 - \tfrac{x^3}{3}\right]_0^2 = 4 - \tfrac{8}{3} = \tfrac{4}{3}.$$

Area between a curve and the y-axis HL

Sometimes it is easier to integrate with respect to y: express x as a function of y and accumulate horizontal slices. This gives the area between the curve and the y-axis.

KEY · IB

$$A = \int_a^b x\, dy$$

EXAMPLE 16.17

Find the area between the curve $x = y^2$ and the y-axis from $y = 0$ to $y = 2$.

$$A = \int_0^2 y^2\,dy = \left[\tfrac{y^3}{3}\right]_0^2 = \tfrac{8}{3}.$$

Integrating in y avoids splitting the region, which would be awkward in x.

YOUR TURN 16.7 Areas Between Curves

13. Find each enclosed area.

(a) Between $y = x$ and $y = x^2$. [3]
(b) Between $y = 4 - x^2$ and the x-axis. [3]
(c) Between $y = 6 - x^2$ and $y = 2$. [3]
(d) Between $x = y^2$ and the y-axis, from $y = 0$ to $y = 3$. [2]

16.8 Volumes of Revolution HL

Rotate a curve about an axis and it sweeps out a solid. To find the volume, slice the solid into thin disks perpendicular to the axis. Each disk is a cylinder of radius y (the curve's height) and thickness dx, with volume $\pi y^2\,dx$. Summing the disks and letting their thickness shrink turns the sum into an integral.

KEY · IB

$$V = \int_a^b \pi y^2\,dx \quad \text{(about the } x\text{-axis)}, \qquad V = \int_a^b \pi x^2\,dy \quad \text{(about the } y\text{-axis)}$$

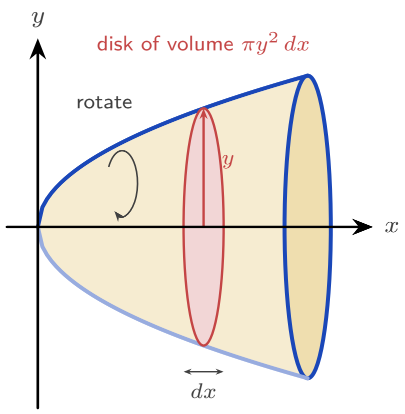

Figure 16.5 Rotating the curve about the x-axis sweeps out a solid. One thin slice, shown in scarlet, is a

disk whose radius is the height y of the curve and whose thickness is dx, so its volume is $\pi y^2\,dx$. The integral adds the disks from one end of the solid to the other.

EXAMPLE 16.18

The curve $y = x^3$ for $0 \le x \le 2$ is rotated about the x-axis. Find the volume.

$$V = \int_0^2 \pi(x^3)^2\,dx = \pi\int_0^2 x^6\,dx = \pi\left[\tfrac{x^7}{7}\right]_0^2 = \frac{128\pi}{7}.$$

Square the function *before* integrating; squaring after is wrong.

EXAMPLE 16.19

The region under $y = x^2$ from $x = 0$ to $x = 1$ is rotated about the y-axis. Find the volume. About the y-axis, slice into disks of radius x and thickness dy, with $x^2 = y$:

$$V = \int_0^1 \pi x^2\,dy = \pi\int_0^1 y\,dy = \pi\left[\tfrac{y^2}{2}\right]_0^1 = \frac{\pi}{2}.$$

The limits are now y-values, and x^2 is rewritten in terms of y.

YOUR TURN 16.8 Volumes of Revolution HL

14. Find each volume of revolution about the x-axis.

(a) $y = x^2$, $0 \le x \le 2$. [3]

(b) $y = \sqrt{x}$, $0 \le x \le 4$. [2]

(c) $y = 2x$, $0 \le x \le 3$. [2]

(d) $y = \dfrac{1}{x}$, $1 \le x \le 3$. [3]

15.

(a) The region under $y = x^2$, $0 \le x \le 2$, is rotated about the y-axis. Find the volume. [3]

(b) Explain why the answer differs from the volume in question 14 part (a), even though the same curve and the same interval are used. [2]

16.9 Kinematics by Integration

In Ch. 15 you differentiated displacement to get velocity, and velocity to get acceleration. Integration reverses those two steps. Integrate acceleration to get velocity, and velocity to get displacement, using an initial condition each time to fix the constant.

KEY · IB

$$\text{displacement} = \int_{t_1}^{t_2} v\,dt, \qquad \text{distance travelled} = \int_{t_1}^{t_2} |v|\,dt$$

Displacement is the signed change in position; distance is the total path length. They differ whenever the particle reverses direction, because backward motion subtracts from displacement but adds to distance. On a velocity–time graph the difference is visible: area below the axis counts as negative for displacement, and as positive for distance.

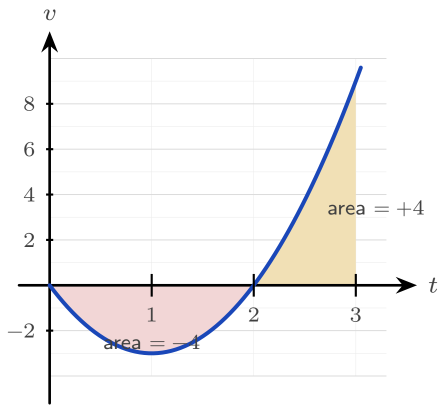

Figure 16.6 The velocity $v = 3t^2 - 6t$ over $0 \le t \le 3$. The particle moves backwards until $t = 2$ and forwards after it. Displacement adds the two signed areas, $-4 + 4 = 0$; distance adds their sizes, $4 + 4 = 8$ m.

EXAMPLE 16.20

A particle has velocity $v(t) = 3t^2 - 6t\,\text{m}\,\text{s}^{-1}$. Find its displacement and the distance travelled over $0 \le t \le 3$.

Displacement integrates v directly:

$$\int_0^3 (3t^2 - 6t)\,dt = \Big[t^3 - 3t^2\Big]_0^3 = 27 - 27 = 0.$$

The particle ends where it began. For distance, note $v = 3t(t-2)$ is negative on $(0, 2)$ and positive on $(2, 3)$, so split:

$$\left|\int_0^2 v\,dt\right| + \int_2^3 v\,dt = |-4| + 4 = 8 \text{ m}.$$

Zero displacement but 8 m travelled: the particle went out 4 m and came back.

YOUR TURN 16.9 Kinematics by Integration

16. A particle moves in a straight line.

(a) Its velocity is $v = 2t + 1$. Find its displacement over $0 \le t \le 3$. [2]

(b) Its acceleration is $a = 6t$ and $v(0) = 1$. Find $v(t)$. [2]

(c) Its velocity is $v = t^2 - 4$. Find its displacement over $0 \le t \le 3$. [2]

(d) Its velocity is $v = 4 - 2t$. Find the distance travelled over $0 \le t \le 4$. [3]

(e) For the particle in part (d), find the displacement over the same interval, and explain why the two answers differ. [3]

Chapter 16 · Integral Calculus

Reflections

IM Long before calculus, Archimedes (3rd century BCE, Syracuse) found the area of a parabolic segment by his "method of exhaustion": filling the region with ever-smaller triangles and summing them. He effectively computed an integral two thousand years before Newton and Leibniz gave the operation a name and a reverse-differentiation shortcut. The idea of accumulation by slicing is older than the algebra we use to carry it out, and his triangles are the ancestors of the rectangles in §16.2.

TOK The proof in §16.2 establishes *that* the area under a curve and the reverse of a derivative are the same calculation. It does not explain *why* the world should be arranged that way. Two ideas invented for different purposes, one geometric and one algebraic, turned out to be one idea.

Does that make the connection a discovery about quantities and their rates, or a consequence of how we chose to define both? Is mathematics invented or uncovered?

RW Integration is summation made continuous, so it appears wherever a total is built from a varying rate. The total charge from a varying current, the work done by a changing force, the area of an irregular plot of land, the probability accumulated under a distribution curve (Ch. 13): each is an integral. Whenever a quantity is the running total of a rate, integration computes it.

EXT Not every function has an antiderivative in terms of familiar functions: $\int e^{-x^2}\,dx$, central to statistics, cannot be written with powers, exponentials, and trigonometric functions. Yet the definite integral still exists as a perfectly real area, and is computed numerically. The reach of "find the area" is wider than the reach of "reverse the derivative."

End-of-Chapter Problems — Chapter 16: Integral Calculus

1. A curve has $\frac{dy}{dx} = 3x^2 - 4x + 1$ and passes through $(1, 2)$.

(a) Find the equation of the curve. [3]

(b) Find the value of y when $x = 2$. [2]

2. Consider $y = x^2 - 4$.

(a) Find the x-intercepts. [2]

(b) Find the area enclosed between the curve and the x-axis. [4]

3. The curves $y = x^2$ and $y = 8 - x^2$ intersect at two points.

(a) Find the coordinates of the intersection points. [3]

(b) Find the area enclosed between the curves. [4]

4.

(a) Using the substitution $u = x^2 + 4$, find $\displaystyle\int \frac{x}{x^2+4}\,dx$. [3]

(b) Hence evaluate $\displaystyle\int_0^2 \frac{x}{x^2+4}\,dx$, giving an exact answer. [3]

5.

(a) Find $\displaystyle\int xe^{2x}\,dx$ using integration by parts. [3]

(b) Evaluate $\displaystyle\int_0^1 xe^{2x}\,dx$, giving an exact answer. [2]

6. The region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is rotated 360° about the x-axis.

(a) Write down an integral for the volume. [2]

(b) Find the volume. [2]

7. A particle starts at the origin with velocity $5\,\mathrm{m\,s^{-1}}$ and has acceleration $a = 6t - 4\,\mathrm{m\,s^{-2}}$.

(a) Find the velocity $v(t)$. [2]

(b) Find the displacement $s(t)$. [2]

(c) Find the displacement when $t = 2$. [2]

8. Consider the region enclosed by $y = x^2$, the y-axis, and $y = 4$.

(a) Find the area of the region. [3]

(b) Find the volume when this region is rotated about the y-axis. [3]

9.

(a) Find $\displaystyle\int \frac{1}{x^2+4}\,dx$. [2]

(b) Evaluate $\displaystyle\int_0^2 \frac{1}{x^2+4}\,dx$, giving an exact answer. [3]

10. The velocity of a particle is $v(t) = 3t^2 - 12t + 9\,\mathrm{m\,s^{-1}}$, $t \geq 0$.

(a) Find the times when the particle is at rest. [2]

(b) Find the total distance travelled in the first 3 seconds. [5]

11. A curve has gradient function $\dfrac{dy}{dx} = 4x - 3$ and passes through the point $(2, 5)$.

(a) Find the equation of the curve. [3]

(b) Write down its y-intercept. [1]

12. Find the following integrals.

(a) $\displaystyle\int e^{3x}\,dx$ [2]

(b) $\displaystyle\int (4x-1)^6\,dx$ [2]

(c) $\displaystyle\int \sin 2x\,dx$ [2]

13. Evaluate $\displaystyle\int_1^4 \left(\frac{2}{\sqrt{x}} + 1\right) dx$. [4]

14. Consider the curve $y = x^2 - 1$ on the interval $0 \leq x \leq 2$.

(a) Find where the curve crosses the x-axis in this interval. [1]

(b) Find the total area enclosed between the curve and the x-axis on the interval. [5]

15. Find the area of the region enclosed by the curve $y = 4x - x^2$ and the line $y = x$. [5]

16. Using the substitution $u = x^2 + 1$, evaluate $\displaystyle\int_0^2 x(x^2+1)^3\,dx$. [5]

17. Evaluate $\displaystyle\int_0^2 \frac{x}{x^2+4}\,dx$, giving your answer in the form $a \ln b$. [5]

18. Find $\displaystyle\int \frac{x+7}{(x-1)(x+3)}\,dx$. [6]

19. Find $\displaystyle\int \frac{1}{x^2 - 4x + 13}\,dx$. [5]

20. Find the following integrals.

(a) $\displaystyle\int x\cos x\,dx$ [3]

(b) $\displaystyle\int x^2 \ln x\,dx$ [4]

21. The region under $y = \sqrt{x}$ from $x = 0$ to $x = 4$ is rotated about the x-axis.

(a) Find the volume generated. [3]

(b) The same curve, written $x = y^2$ for $0 \le y \le 2$, is rotated about the y-axis. Find the volume generated. [4]

22. A particle moves with velocity $v(t) = t^2 - 4t + 3\,\mathrm{m\,s^{-1}}$ for $0 \le t \le 4$.

(a) Find the displacement of the particle over the interval. [3]

(b) Find the total distance travelled. [5]

23. The area under $y = x^2 + 1$ from $x = 0$ to $x = 3$ is to be estimated by three rectangles of width 1.

(a) Find the estimate obtained by taking the height of each rectangle at the left of its strip. [3]

(b) Find the estimate obtained by taking the height at the right. [2]

(c) Evaluate $\displaystyle\int_0^3 (x^2 + 1)\,dx$ and verify that it lies between the two estimates. [3]

(d) State, with a reason, whether the left estimate would increase or decrease if six rectangles were used instead. [2]

24. The curve $y = x^2$ and the line $y = 2 - x$ enclose a region R.

(a) Find the x-coordinates of the two points of intersection. [3]

(b) Sketch the curve and the line, and shade R. [2]

(c) Show that the area of R is $\dfrac{9}{2}$. [5]

(d) Explain why integrating $y_{\text{curve}} - y_{\text{line}}$ instead would give the wrong sign. [2]

25. This question uses integration by parts throughout.

(a) Find $\displaystyle\int x\ln x\,dx$. [4]

(b) Hence evaluate $\displaystyle\int_1^e x\ln x\,dx$, giving your answer in terms of e. [3]

(c) Evaluate $\displaystyle\int_0^{\pi/2} x\cos x\,dx$. [5]

(d) In part (c), state which factor you took as u, and explain what would go wrong with the other choice. [2]

26. Consider the curve $y = x^3$ for $0 \le x \le 1$.

(a) The region between the curve and the x-axis is rotated 360° about the x-axis. Show that the volume is $\dfrac{\pi}{7}$. [4]

(b) The region between the curve and the y-axis is rotated 360° about the y-axis. Find this volume, giving an exact answer. [5]

(c) State which of the two solids has the greater volume, and explain the answer by describing the two regions. [3]

27. A particle moves in a straight line with acceleration $a = 6 - 2t\,\mathrm{m\,s^{-2}}$, and is at rest at $t = 0$.

(a) Show that $v = 6t - t^2$. [3]

(b) Find the time at which the particle is next at rest. [2]

(c) Find the maximum velocity, and the time at which it occurs. [3]

(d) Find the distance travelled from $t = 0$ until the particle is next at rest. [4]

(e) Explain why, for this motion, the distance travelled equals the displacement. [2]

28. Each of these integrals needs a different method. In each part, name the method before using it.

(a) $\displaystyle\int \frac{x^2+1}{x+1}\,dx$ [5]

(b) $\displaystyle\int_1^2 \frac{2x-1}{x^2-x+1}\,dx$ [4]

(c) $\displaystyle\int_0^{\pi/2} \sin x \cos^2 x\,dx$ [4]

(d) $\displaystyle\int \frac{1}{x^2-4x+13}\,dx$ [4]

Challenge Questions

29. **Solids you already know.** Every volume formula from Ch. 8 can be recovered by rotating a curve. This question derives three of them.

(a) The line $y = r$ for $0 \le x \le h$ is rotated about the x-axis. Show that the volume is $\pi r^2 h$, and name the solid. [3]

(b) The line $y = \dfrac{r}{h}x$ for $0 \le x \le h$ is rotated about the x-axis. Show that the volume is $\frac{1}{3}\pi r^2 h$. [4]

(c) The semicircle $y = \sqrt{r^2 - x^2}$ for $-r \le x \le r$ is rotated about the x-axis. Show that the volume is $\frac{4}{3}\pi r^3$. [5]

(d) The cone in part (b) has one third of the volume of the cylinder in part (a). Explain where the factor $\frac{1}{3}$ comes from in the integration. [3]

30. When parts goes in a circle. Some integrals return to where they started, and that is what solves them.

(a) Let $I = \int e^x \cos x \, dx$. Apply integration by parts twice, and show that $I = \frac{1}{2}e^x(\sin x + \cos x) + C$. [6]

(b) Explain why choosing $u = e^x$ at the first step and $u = \cos x$ at the second would not have worked. [3]

(c) Find $\int x^2 e^x \, dx$ by applying parts twice, and explain why this one terminates while part (a) does not. [5]

(d) Predict, without calculating, how many rounds of parts $\int x^4 e^x \, dx$ requires. Justify your answer. [2]

31. A reduction formula. Let $I_n = \int_0^{\pi/2} \sin^n x \, dx$ for $n \geq 0$.

(a) Evaluate I_0 and I_1 directly. [3]

(b) Writing $\sin^n x = \sin^{n-1} x \cdot \sin x$ and using integration by parts, show that

$$I_n = \frac{n-1}{n} I_{n-2}, \qquad n \geq 2.$$

[6]

(c) Use the formula to find I_2, I_3, I_4 and I_5 exactly. [4]

(d) Explain why I_n involves π when n is even but not when n is odd. [3]